

These values of coefficients are enough to reconstruct the full dependence:

$$\begin{aligned} A_{a,b,c} &= 2(a+b+c) + 3 \\ C_{a,b|c,d} &= 3(2a+1)(2b+1) \\ G_a &= (2a+1)(2a^2+2a+1) \end{aligned} \quad (132)$$

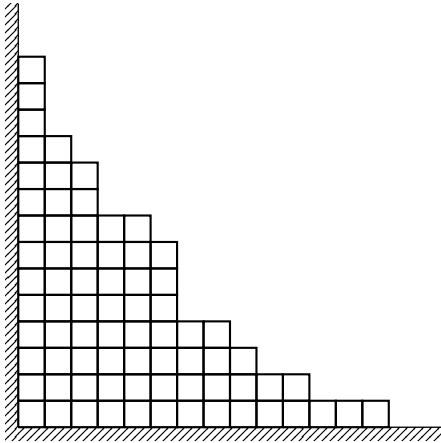
Now we are able to write the full Hamiltonian for SIB polynomials:

$$\begin{aligned} \hat{\mathcal{H}} &= \sum_{a,b,c=0}^{\infty} (2(a+b+c) + 3) \tau_{a+b+c+1} \frac{\partial^3}{\partial \tau_a \partial \tau_b \partial \tau_c} + \sum_{a+b=c+d} 3(2a+1)(2b+1) \tau_a \tau_b \frac{\partial^2}{\partial \tau_c \partial \tau_d} \\ &\quad \sum_{a,b,c=0}^{\infty} 4(2a+1)(2b+1)(2c+1) \tau_a \tau_b \tau_c \frac{\partial}{\partial \tau_{a+b+c+1}} + \sum_{a=0}^{\infty} (2a+1)(2a^2+2a+1) \tau_a \frac{\partial}{\partial \tau_a} \end{aligned} \quad (133)$$

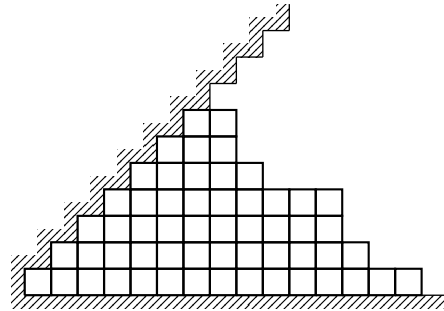
One can check that eigenvalues of the above operator are positive integers and obey formula (130). This Hamiltonian itself does not define SIB basis due to degenerate spectrum, for example $\varepsilon_{[8,7]} = 8^3 + 7^3 = 855 = 9^3 + 5^3 + 1^3 = \varepsilon_{[9,5,1]}$. However, with adding/removing rules the basis is fixed completely.

3.5 Action of Virasoro operators on SIB_λ polynomials

Motivated by the action of time-variables and Virasoro operators on SIB polynomials that we discuss in this section it is instructive to slightly change notation of diagrams DP - distinct partitions. For this purpose we represent a Young diagram (turned upside down) as a way of tight packing of square boxes in a corner of a room. In this setting distinct partitions (DP) are in one-to-one correspondence with possible ways of tight packing of square boxes under a ladder. We demonstrate examples of these diagrams, left picture correspond to usual Young diagram, right one correspond to distinct partition:



(a) Usual partition



(a) Distinct partition

(134)

Form now on we represent distinct partitions as Young diagrams under the ladder. With these notations the action of τ_0 and $\frac{\partial}{\partial \tau_0}$ have the following from:

$$\tau_0 \cdot \text{SIB}_\lambda = \sum_{\square \in \text{AddDP}(\lambda)} \text{SIB}_{\lambda+\square} \quad (135)$$